

GJØVIK UNIVERSITY COLLEGE

BACHELOR THESIS

Snowman

A complete, effective and secure rule management system for SNORT®

Authors:
Thomas NYHEIM
Eirik SKOGSTAD
Eigil OBRESTAD

Supervisor:
Slobodan PETROVIC

March 16, 2014

Preface

This is the preface.

Glossary

MD5 A cryptographic hash function producing a 128-bit hash value.

URL Uniform Resource Locator.

Contents

Preface . . . . . **i**
Glossary . . . . . **ii**
Contents . . . . . **iii**
List of Figures . . . . . **iv**
List of Tables . . . . . **v**
1 INTRODUCTION **1**
2 EXISTING SOLUTIONS **2**
3 REQUIREMENTS SPECIFICATION **3**
4 DESIGN **4**
5 IMPLEMENTATION **5**
5.1 Frameworks 5
5.2 Concepts of Snort® 5
5.2.1 Rules 5
5.2.2 Generators 6
5.2.3 Reference Types 6
5.2.4 Event filters 6
5.3 Core module 6
5.4 Update module 6
5.4.1 Sources 6
5.4.2 Files 6
5.4.3 Running an update 7
5.4.4 Rule parsing 7
5.4.5 Classification, Generator and Reference Type parsing 8
5.4.6 Logging 8
5.5 Graphical User Interface 8
5.5.1 Structure 8
5.5.2 Design 8
5.5.3 Features 9
5.5.4 Rules/All rules page 9
6 OPTIMIZATION **11**
6.1 Update 11
7 TESTING **12**
8 RESULTS **13**
9 ANALYSIS OF RESULTS **14**
10 CONCLUSIONS **15**
Bibliography **16**

List of Figures

1 GUI Structure 9
2 Rules/All rules page 10

List of Tables

1 Determining whether a rule is enabled or disabled on a sensor. 5
2 Rule options used in this system. 7

1 INTRODUCTION

Følgende informasjon kan tas med her: - Organisering av rapporten og kort beskrivelse av sammenhengen mellom kapitlene. - Fullstendig definering av oppgaven med utdypning av hva oppgaven går ut på. Eksempelvis presiseres mål, problemområde, omfang og avgrensning. - Henvisning til eventuelle tidligere arbeider om samme emne. - Målgruppe for rapporten, jfr. mål- og mottakeranalysen. - Studentenes faglige bakgrunn. - Prosjektgruppens valgte arbeidsformer, f.eks. samtaler og intervjuer med kilder, reiser, bedriftsbesøk, litteraturstudier, laboratorieforsøk, modellbygging og testing. - Terminologibruk. Er eksempelvis faguttrykk brukt i en spesiell betydning? Innledningen skal ikke inneholde noe om det som er gjort i oppgaven og oppnådde resultater.

2 EXISTING SOLUTIONS

Her tenker jeg at vi tar med hva som fins av programvare fra før og peker påutfordringer i disse løesningene.

3 REQUIREMENTS SPECIFICATION

Med bistand fra oppdragsgiver skal det gjennomføres en analyse av hvilke krav som stilles til det nye systemet. Arbeidet skal resultere i et strukturert og godt kravspesifikasjonsdokument som definerer de funksjonelle og operasjonelle kravene til systemet. Dokumentet legges i sin helhet inn i prosjektrapporten under vedlegg. Det skal kort redegjøres for metodebruk og fremgangsmåte i arbeidet med å komme frem til spesifikasjonen.

4 DESIGN

Utfyllende med grunngitt valg av metoder, rammeverk etc. Hovedtrekk fra design av system beskrives. Fullstendig designdokument bør være vedlegg til prosjektrapporten, mens dette punktet omhandler diskusjon av sentrale forhold. Ulike alternative design som studentene vurderte bør beskrives og valg av alternativ begrunnes.

5 IMPLEMENTATION

This chapter will focus on the technical implementation of the software.

5.1 Frameworks

To simplify (db, web, code) this project uses the Django framework.

5.2 Concepts of Snort®

This section will give a brief introduction to the basic concepts of Snort®, and discuss these in relation to features made in this project.

5.2.1 Rules

A rule is basically a set of options that define ways of detecting, classifying and reporting a potential attack or threat on the network[1]. There are several types of options, but only the general rule options are relevant for this project. The general rule options provide information about the rule[1] (i.e. revision number, alert message). Other types of options, such as payload options which are actively used in detection, are not relevant for the management of rules and are thus not considered in this project.

A rule can either be enabled or disabled in the detection system, meaning that the rule will only have an effect on detection when it is enabled. Since this project is dealing with multiple sensors connected to one central database, this functionality has been expanded so that users can keep a rule enabled on one sensor and disabled on another. In addition, the user can also set an enabled/disabled flag on rulesets.

A rule *modifier* is created for each rule in a ruleset when the ruleset is added to a sensor. The modifier dictates whether the rule will be enabled or not on the given sensor, and by default it will assume the original value of the rule. This means that the original value is only used to determine the default value of the modifier. A modifier can also be configured to "follow the ruleset", in which case the rule will be enabled only when the ruleset is enabled, and vice versa. See table 1 for the effects of the modifier and ruleset options.

[1] A rule actually consists of a header and options, rtfm.[1]

Modifier	Ruleset	Active on sensor?
Enabled	*any*	Yes
Disabled	*any*	No
Ruleset	Enabled	Yes
Ruleset	Disabled	No

Table 1: Determining whether a rule is enabled or disabled on a sensor.

5.2.2 Generators

5.2.3 Reference Types

5.2.4 Event filters

Threshold deprecated. Detection filters.

5.3 Core module

Lorem, ipsum?

5.4 Update module

The purpose of the Update module is to fetch and read rules and configuration files from external sources, and populate the database with this information. Effectiveness in updates is a key issue, and a number of deliberate optimizations for performing updates, with the goal of saving time by omitting unnecessary operations, are explained in the following sections.

The term Update is used to refer to the actual module, including all its classes and functions. To avoid confusion with the module, the term *update-package* is specifically used to refer to update data coming from external sources.

This section will define an example rule to be used throughout subsequent sections and the remainder of the chapter, to illustrate the lifespan of a rule in the system.

Word: rulestring.

5.4.1 Sources

A *source* is used by Update to determine where the update-package is located. Normally this is on a remote site, and the source contains the URL pointing to the location of the update-package. A source can be configured to update at regular intervals, or the user can run an update manually on a source. To make the update process more effective, Update makes use of an MD5 URL which points to a file containing the MD5 hash of the update-package. This hash is stored for every source (if available) and is compared to any pre-existing hash to verify if the system is up to date or not. Update will only download and process an update-package when it encounters a new hash.

5.4.2 Files

An update-package may contain various files depending on the source it comes from. This section will describe files supported by Snowman.

Rule files

A rule file is any file carrying the *.rules* file extension. These files contain raw rulestrings to be processed by Snort. A rulestring is basically a semicolon-separated list of rule options. Specific rule options used in this system are listed in table 2.

A badly formatted rulestring given to Snort will cause a crash, and Snort must be restarted.

Classification, Generator and Reference Configuration files

The classification files are identified by their name *classification.config* and contain rule classification strings that are used by Snort to categorize event data. A classification file will normally appear once per update-package, and rarely changes between updates[2].

[2]https://www.snort.org/vrt/snort-conf-configurations/

Data field	Description	Presence
sid	Unique identifier for a rule.	required
rev	Rule revision number.	required
msg	Notification text when rule triggers.	required
classtype	Connects the rule with a specific class of attacks.	required
gid	Generator ID to identify which subsystem generates the event where this rule fires.	optional
ruleset	The ruleset this rule belongs to.	optional
reference	Reference to external resource with information about this attack.	optional

Note: The *presence* column indicates whether a field is required or not by the parser in this system. Information is gathered from the Snort Users Manual [1](ch 3.4).

Table 2: Rule options used in this system.

Sid message files

5.4.3 Running an update

As an example scenario, consider a rule (name, figure reference, source, ruleset etc.). An update is triggered on the source at the predefined time

The update is cancelled if the source is already up to date; the MD5-hashes of the latest update in the database, if any, and the latest update on the remote site are compared, and the process is terminated if these hashes are identical. Otherwise, the update is downloaded to a temporary directory and the update process continues. A file can be a tar-archive containing an arbitrary number of rulesets and configuration files, or it can be a plain text file containing Snort configuration (?).

The tar-archive is extracted, and the update module recursively parses each file in each folder. In most cases the update-package will contain only minor changes in relation to its predecessor, e.g. only one file is different in the latest package, and parsing all files that has not changed would therefore be a waste. To address this issue and save processing time, the name, path and MD5-hash of every file is stored in the database per source. Every time the update module parses a file, it first creates a hash of the file and compares this with the respective file's hash in the database. If the file is present in the database and the two hashes coincide, the file is considered *not new* and is skipped. The following sections will elaborate on the parsing of files based on file type.

parse files with regex.

5.4.4 Rule parsing

Some options can be defined both in the rulestring (inline) and in a separate file (standalone).

If any properties classified as *required* for a rule (table 2) are missing...

Non-existing rule sets, rule classes and generators are stored in the database

After the initial parsing of the rulestring, the parser checks if the rule already exists in the database, and if so whether the incoming revision number is higher than the latest

revision of the pre-existing rule. The parsing will continue only in the event of a new rule or a newer revision of an existing rule.

rule moved to new ruleset.

Rule sets

A rule always belogs to a ruleset which can be specified in the raw rulestring. If the parser does not find this attribute in the rulestring, the filename is considered to be the ruleset name.

Threshold

A rule threshold defines a limit and/or threshold for the triggering of a specific rule, and can exist as an inline or standalone option. Rule thresholds are deprecated, and support for this functionality will cease in a future version of Snort®, where it is replaced by *detection filters* and *event filters* (see below) [1]. To maintain compatibility with current rules, Snowman supports reading inline and standalone threshold definitions.

Detection filters

Please explain..

Event filters

Please explain..

Suppression

Always standalone.

5.4.5 Classification, Generator and Reference Type parsing

These files contain a sequence of their respective entities (section 5.4.2), and are all parsed in the same manner. If a given entity is not already present in the database, it is created and stored. If the database already contains the entity, the existing entry in the database is overwritten.

Rule Message files

5.4.6 Logging

5.5 Graphical User Interface

The Graphical User Interface (GUI) is a template- and AJAX-based (desc/footnote?) Web application which uses Django's (desc) template system to implement the View part of the MVC-principle(ref). Each webpage is a template populated through Python-code before it is delivered to the Web server and ultimately the requesting Web browser.

Additional data that needs to be sent to a Web page, e.g. the user interacts with the GUI, is requested through JavaScript(desc) and AJAX-calls and then gets added to the already loaded DOM(desc) in the web browser.

5.5.1 Structure

The pages of the GUI is organized in the following structure:

5.5.2 Design

The GUI is designed with simplicity in mind, which is achieved by using a front-end framework called Bootstrap(ref). This framework provides easy-to-use and ready-made design elements for HTML and CSS(desc), which has been crucial for saving time and energy in the project. Consider the example job of creating a <table> element and spend-

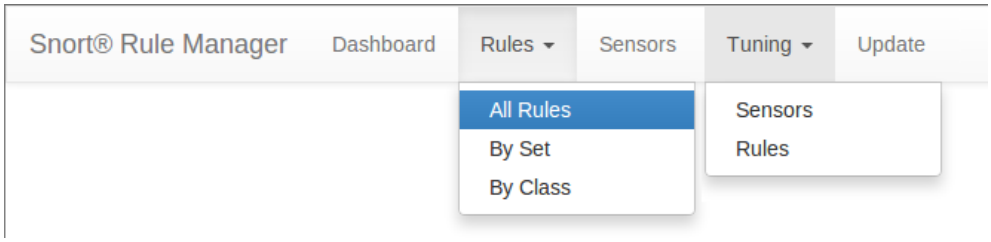


Figure 1: GUI Structure

ing hours setting up and tweaking the CSS for the said table element, we could just add <table class='**table table-bordered**'> and get the table formatted nicely with colors and fonts that are typographically matching.

5.5.3 Features

Content Prefetching

One of the main features of our GUI design is our usage of prefetching to speed up loading. Instead of loading absolutely everything into the web browsers DOM, like our predecessor BHTB did, we instead initially only load the first page or page elements. This significantly reduces loading times.

We then silently in the background load in any element the user might want to see next, through AJAX calls. When the user then interacts with the GUI, the next elements are shown to the user. The next-next elements are also loaded through AJAX calls so that they are ready.

This ensures that the user always has immidiate response from any interaction done with the GUI, while keeping loading times to a bare minimum.

5.5.4 Rules/All rules page

This page serves as the complete list of all rules currently residing in the central database. Because the central database can literally contain hundreds of thousands of rules, this list is paginated. Pages are loaded using AJAX calls to prefetch them as needed.

Each row in the list can be clicked on, which expands an area containing additional information about the rule in question.

We anticipate that the main use of this page is to utilize the search function to find a specific rule or rules and then either retrieve information about them or tune them.

Rules can be tuned by checking the checkbox associated with the rule in question and then press the button that represent the desired function in the box to the left.

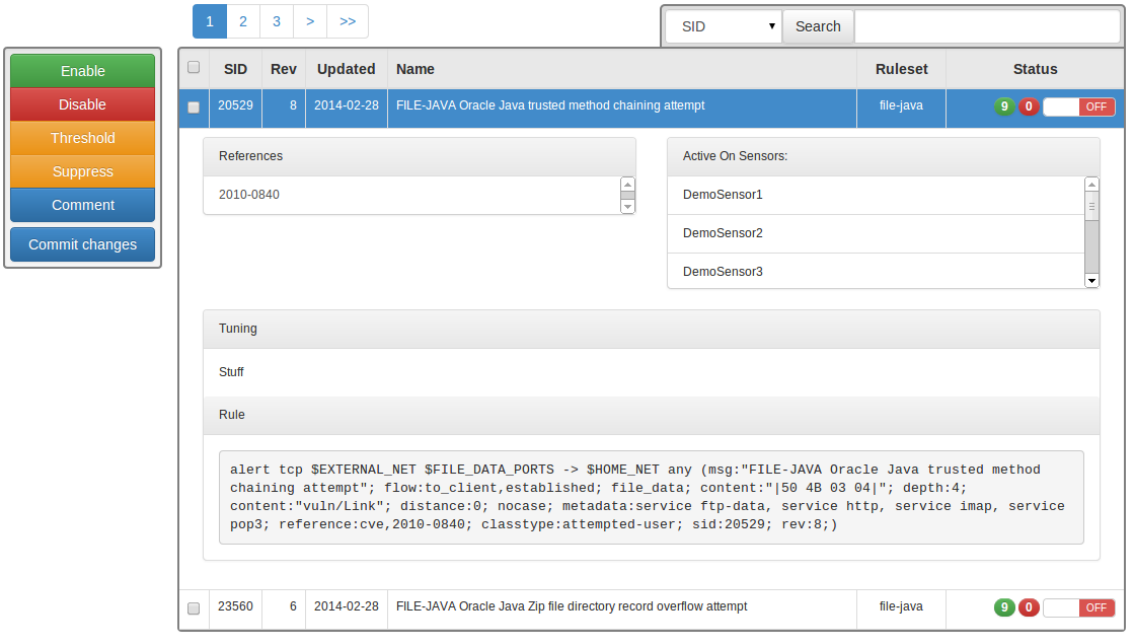


Figure 2: Rules/All rules page

6 OPTIMIZATION

6.1 Update

7 TESTING

Tilstrekkelig beskrivelse til at forsøk og undersøkelser kan reproduseres og etterprøves: - Forsøksbetingelser, og hvordan disse stemmer med prinsipper beskrevet i kapittel 2. - Praktisk fremgangsmåte. - Analysemetode (oftest i vedlegg).

Her beskrives hvordan opplegget rundt testing av applikasjonen har vært. Dette omfatter planlegging, valg av teststrategier, gjennomføring av tester og beskrivelser av sentrale feil funnet under testing, samt måten feil ble rettet opp på.

8 RESULTS

De faktiske observasjoner og data fra forsøk (f.eks. målinger) og undersøkelser (f.eks. utspørringer) presenteres på en oversiktlig måte ved tabeller og/eller diagrammer. Bruk skjønn for evt. plassering i vedlegg. Ved bruk av diagrammer skal det gis beskrivende tekst til grafer. Resultatene skal kort kommenteres, men ikke diskuteres. Viktige resultater gjengis kortfattet under ”Sammendrag”. En oversikt over tabeller og diagrammer med nummer og titler kan plasseres etter innholdsfortegnelsen.

9 ANALYSIS OF RESULTS

Diskusjon rund resultatene. Det skal gjøres en evaluering av hvorvidtvalgt løsning. Avvik drøftes. det er samsvar mellom kravspesifikasjonen satt opp og

10 CONCLUSIONS

De slutninger som kan trekkes fra resultatene og deres diskusjon skal være logisk og nøkternt oppbygd. Etterrettelighet er vesentlig. Konklusjoner skal fremkomme klart og uten tvetydigheter. Eventuell usikkerhet påpekes. Grunnlaget for slik tvil skal redegjøres under kapittel 6a/6b. De viktigste konklusjoner blir presentert i kortfattet versjon under ”Sammendrag”. Prosjektgruppen bør ta stilling til hva de faglige resultatene kan brukes til, hva gruppen faglig har lært av dette arbeidet, hva som evt. kunne vært gjort annerledes ved en gjentagelse av utviklingsarbeidet, og hvordan eventuelt videre arbeid rundt samme eller beslektet tema bør gjøres. Gruppen bør også kommentere den subjektive opplevelse av hovedprosjektet, og hvordan gruppeprosessen har vært. Kort sagt: Egenevaluering av gruppearbeidet.

Bibliography

[1] Roesch, M. 2012. *Snort Users Manual 2.9.3*. The Snort Project.